Title

File Handling and Pattern Frequency Count in Java

Objective

To write a Java program to write text into a file named `sample.txt`, read the file, and count the occurrences of the patterns `pe` and `pi`.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

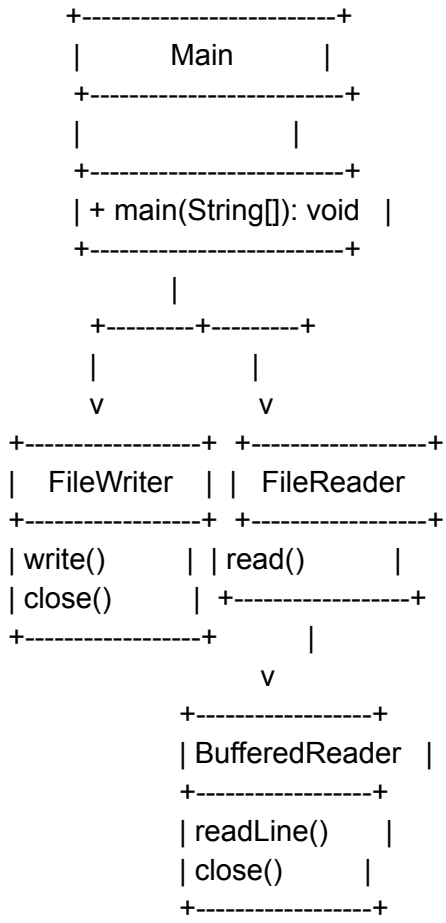
Software Requirements

- Java JDK
- VS Code / Eclipse / IntelliJ IDEA
- Windows/Linux operating system

Algorithm / Logic Description

1. Start the program.
2. Declare a string containing the given text.
3. Create a `FileWriter` object to write the text into `sample.txt`.
4. Write the text into the file and close the writer.
5. Create a `FileReader` and `BufferedReader` to read the file.
6. Initialize two counters, `peCount` and `piCount`, to zero.
7. Read the file line by line.
8. Traverse each line and extract every two-character substring.
9. If the substring matches `pe` (ignoring case), increment `peCount`.
10. If the substring matches `pi` (ignoring case), increment `piCount`.
11. Close the reader after reading the file.
12. Display the occurrence counts of `pe` and `pi`.
13. Handle any file-related exceptions.
14. Stop the program.

UML Class Diagram



Source Code

```
import java.io.*;

public class Main {
    public static void main(String[] args) {
        String text = "Peter Piper picked a peck of pickled peppers\n" +
                      "A peck of pickled peppers Peter Piper picked\n" +
                      "If Peter Piper picked a peck of pickled peppers\n" +
                      "Where's the peck of pickled peppers Peter Piper picked?";

        int peCount = 0, piCount = 0;

        try {
            FileWriter fw = new FileWriter("sample.txt");
            fw.write(text);
            fw.close();
```

```
        FileReader fr = new FileReader("sample.txt");
        BufferedReader br = new BufferedReader(fr);

        String line;

        while ((line = br.readLine()) != null) {
            for (int i = 0; i < line.length() - 1; i++) {
                String pattern = line.substring(i, i + 2);

                if (pattern.equalsIgnoreCase("pe"))
                    peCount++;

                if (pattern.equalsIgnoreCase("pi"))
                    piCount++;
            }
        }

        br.close();

        System.out.println("'pe' - no of occurrences - " + peCount);
        System.out.println("'pi' - no of occurrences - " + piCount);

    } catch (IOException e) {
        System.out.println("File error: " + e.getMessage());
    }
  }
}
```

.

Compilation and Execution Commands

Compile:

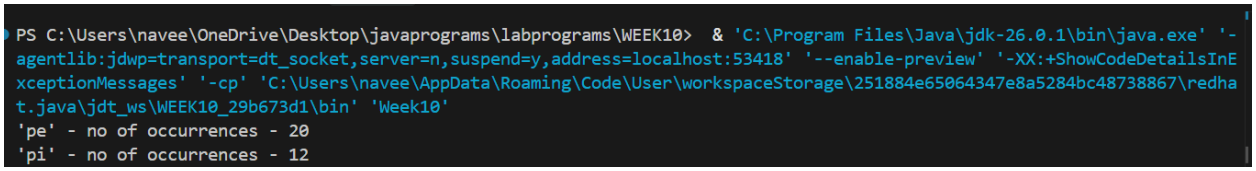
javac Main.java

Execute:

java Main

Output Screen

'pe' - no of occurrences - 20
'pi' - no of occurrences - 12



Result

The Java program was implemented to write text into `sample.txt`, read its contents, and count the occurrences of the patterns `pe` and `pi` using Java file-handling classes.